

Design and Optimization of OTA-C Filters with Shared CMFB and Output Stages: Performance, Power and Area Analysis

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Abstract—This paper presents an optimized approach to designing OTA-C filters with a focus on reducing the number of components while maintaining or improving performance. By employing shared sub-blocks within the transconductance amplifier architecture, the proposed method minimizes power consumption and circuit area, ensuring efficient integration in modern electronic systems. The paper also discusses the theoretical basis for the design, implementation details, and performance evaluation of the optimized filters. The results demonstrate significant improvements in resource efficiency and scalability compared to conventional designs.

Keywords—OTA, OTA-C filters, optimisation, simulation, design, CMFB, shared sub-blocks, performance enhancement

I. INTRODUCTION

Despite significant advancements in technology, analog filters continue to play a vital role in numerous applications, ranging from communication systems to signal processing. One of the key factors determining the practicality of electronic circuits is power consumption. Reducing the energy usage of a circuit not only minimizes the size of the device but also extends its operational time on a single battery charge.

In recent years, significant advancements have been made in the design of analog filters, particularly in optimizing their performance while minimizing power consumption and component count. For instance, Kumngern et al. presented a fifth-order Butterworth low-pass filter tailored for ECG signal acquisition, utilizing a multiple-input operational transconductance amplifier (MI-OTA) [1]. Their approach demonstrated exceptional efficiency, achieving a power consumption of 34.65 nW while maintaining a wide dynamic range and high reliability through PVT and Monte Carlo analyses [1].

Similarly, Kulej et al. introduced a fully differential multiple-input OTA operating at ultra-low power levels, which enabled the realization of a fifth-order Chebyshev low-pass filter for bio-signal processing [2]. By leveraging bulk-driven MOS transistors in the subthreshold region, they achieved a power consumption of just 60 nW, showcasing the potential of MI-OTA in low-power filter applications [2].

Furthermore, Wyszynski and Schaumann highlighted the advantages of employing multiple-input OTAs in filter design, particularly in reducing the number of components, silicon area, and power dissipation [3]. Their findings emphasized the scalability of MI-OTA-based designs and their impact on enhancing the overall efficiency of OTA-C filters [3].

As shown above, various approaches to optimizing analog filters have lately been explored in the literature. These methods typically focus on reducing power consumption, improving frequency characteristics, or minimizing the number of required components. This paper builds on these concepts by proposing a novel design method for OTA-C filters. The approach involves replacing the integrated OTA amplifier with a version composed of sub-blocks, which allows for the elimination of redundant elements in higher-order OTA-C filters (above the second order). This study demonstrates that such a solution allows for the realization of filters with comparable noise performance and signal distortion, without any degradation in these characteristics, while significantly reducing the number of required MOS transistors and power consumption.

II. DESIGN OF OTA AND CMFB

A. Operational Transconductance Amplifier

To present the idea, the starting point is a simple, folded-cascode OTA, although more sophisticated OTAs are also applicable. Its structure was modified to allow it to be divided into two main sub-circuits: the input stage and the output stage. The input stage is responsible for converting the input voltage into current, while the output stage ensures a high output resistance of the amplifier [4][6]. The structure of the described circuit is shown in Figure 1.

The dimensions of the transistors and their operating points were selected to ensure that, with zero input signal, the

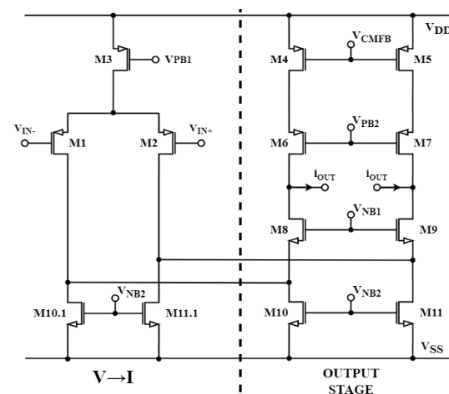
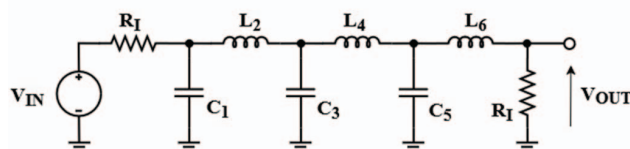


Fig. 1. The designed OTA structure with division into two sub-blocks.

B. Common Mode Feedback Circuit

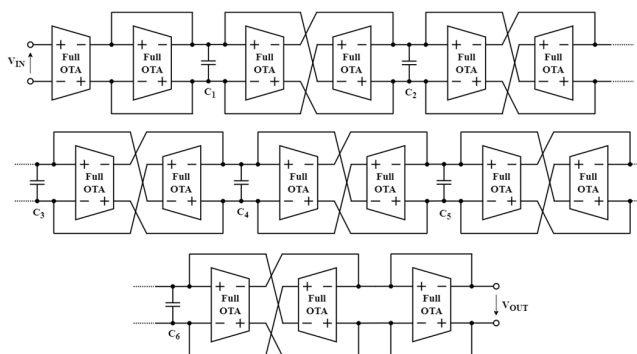
III. FILTER DESIGN

The proposed filter design is based on a RLC ladder shown in Fig. 4 [5]. A 6th order low-pass Butterworth filter was chosen as a representative example due to its wide applicability and the balance of parameters and complexity. The 3-decibel frequency was set to 4.5 MHz as it best fits the OTAs capabilities and is used in real applications, most importantly in analogue audiovisual signals.



Two OTA-C filter designs based on the RLC ladder above were chosen. One was obtained from a direct simulation of the RLC prototype, while the other is based on the signal flow simulation in the aforementioned RLC ladder. Both proposed filters were designed in three versions. The base variants were made from full OTAs, each containing the input stage, output stage, and the CMFB Circuit. Then, in the first approach, we introduced a design with shared CMFB circuits for each filter node, while maintaining OTAs composed of the input and output stages. Lastly, a filter incorporating shared CMFB circuits and output stages was developed, while OTAs from the previous step were reduced to only input stages. The schematics and details of these designs are shown in the paragraphs below.

The operating principle of this filter is based on simulating the filter's passive components. It directly implements the RLC ladder network shown in Fig. 4. To simulate the inductor, a gyrator topology with a capacitor was employed [5]. Fig. 5 shows the realization of this filter developed using full OTAs that integrate CMFB circuits.



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C. Signal Flow OTA-C filter

The concept of the signal flow OTA-C filter's operation is based on the RLC ladder shown in Fig. 4. However, this filter structure is focused on the simulation of the flow of signals in the previously mentioned ladder and its principle of operation [5]. This approach, similarly to the ladder-based filter, also allows for using solely capacitors as passive components. The schematic diagram showing this type of filter is presented in Fig. 6.

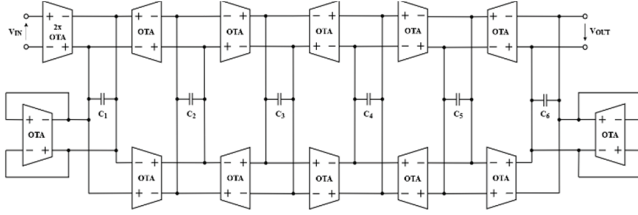


Fig. 6. Signal flow based OTA-C filter composed of full OTA circuits.

D. Optimisation method

The optimization technique applied to the designed filters is based on the division of an OTA into three sub-blocks: input stage, output stage and CMFB. This approach allows for flexible use of these sub-blocks as they can be connected separately.

The first step that was taken builds upon the fact that a CMFB circuit's function is to stabilize the average voltage in the OTA's output node. As its operation is to affect the whole output node, it is possible to use only one such circuit for each of the nodes, where multiple OTAs' outputs meet. Therefore, in this setup the OTAs were cut down only to input and output stages while the CMFB circuit was separated from it. The ladder-based and signal-flow based filters designed with this approach are shown in Fig. 7 and Fig. 8 respectively.

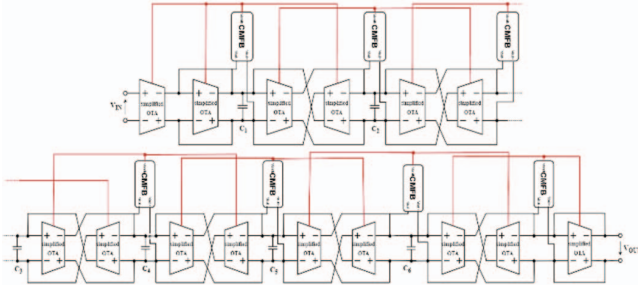


Fig. 7. Ladder based OTA-C filter composed of OTAs with shared CMFB circuits.

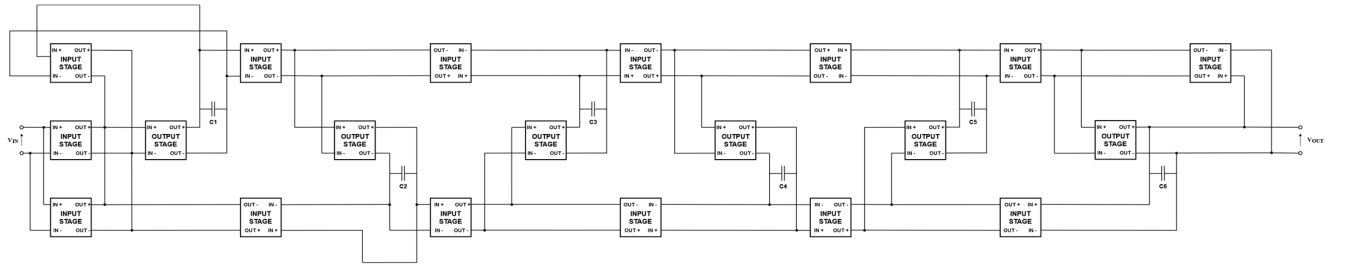


Fig. 10. Signal flow based OTA-C filter composed of OTAs with shared output stages and CMFB circuits.

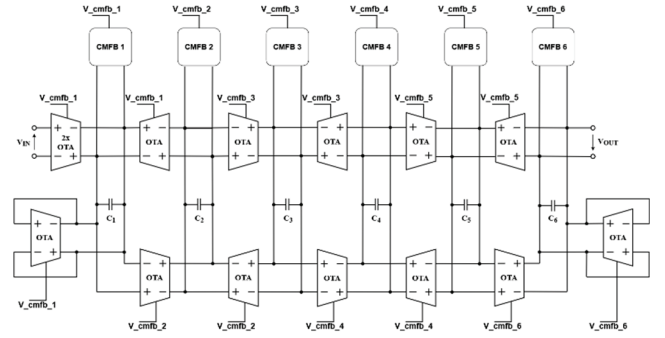


Fig. 8. Signal flow based OTA-C filter composed of OTAs with shared CMFB circuits.

The final filter structure was achieved by extending the method above. The introduced improvements result from the observation that the output stage plays mostly the role in output signal conditioning. It means that multiple input stages can be connected to one output stage, and still work the same as two full OTAs. Therefore, in this approach the OTAs were cut down only to input stages while the output stage and CMFB circuits were separated from it and combined into an output block. Then, the filters were designed with the output blocks shared for every node, in which the OTAs outputs were connected. The ladder-based and signal-flow based filters designed with this approach are shown in Fig. 9 and Fig. 10 respectively.

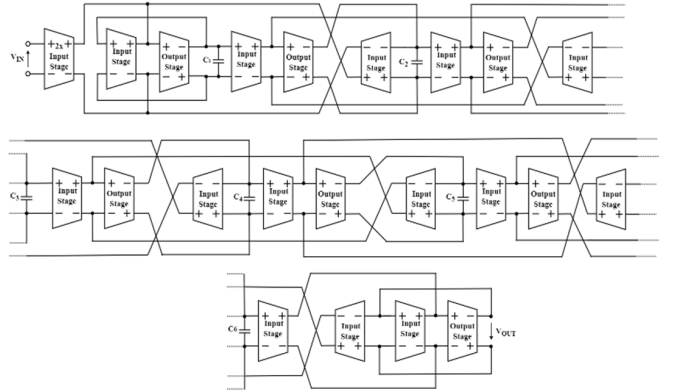


Fig. 9. Ladder based OTA-C filter composed of OTAs with shared output stages and CMFB circuits.

E. Encountered problems

One of the significant challenges identified during the design process involved the initial transconductance (g_m) value of the OTA, which was originally set to $10 \mu\text{S}$. While this value was sufficient for the basic operational requirements of the amplifier, it proved inadequate for practical implementation in OTA-C filters targeting a 3-dB cutoff frequency of 4.5 MHz. Specifically, the low g_m value resulted in capacitor sizes that were too small for accurate fabrication in integrated circuit technology, posing a significant limitation in achieving the desired performance.

To resolve this issue, the g_m of the OTA was increased to $20 \mu\text{S}$. This adjustment allowed for larger capacitor values, ensuring that the filters could be implemented with the necessary precision while maintaining stability and performance. However, this change required a reassessment of other design parameters to accommodate the increased transconductance.

Another challenge encountered during the design process was the reduction in effective transconductance observed in configurations utilizing shared output stages in the OTA. Simulations comparing designs with full OTAs and those with multiple input stages sharing a single output stage revealed a noticeable g_m decrease of approximately $3.6 \mu\text{S}$ when three input blocks were connected in parallel. This reduction becomes more pronounced as the number of input blocks connected in parallel increases. This issue primarily stems from the parallel connection of input stages, which introduces a reduction in the output impedance of the input blocks.

The reduced output impedance leads to a portion of the output current from the differential pairs flowing back through the resistance, rather than being fully directed to the output stage. Consequently, the effective transconductance decreases with an increasing number of input stages connected to a shared output. This phenomenon necessitates adjustments in the design process to mitigate its impact on filter performance.

To address this issue, a careful re-calibration of the capacitors used in the OTA-C filters was required. Specifically, the capacitance values had to be decreased to compensate for the reduced transconductance, ensuring that the filters maintained their desired frequency and amplitude characteristics despite the observed transconductance drop. This adjustment highlighted the need for precise parameter tuning when designing filters with shared output stages, to balance power consumption, area optimization, and overall performance.

IV. RESULTS

In order to provide data concerning the filters' performance, area, and power consumption, appropriate simulations were performed. They were based on models provided with the GDPK 45nm technology, the transistors used worked with a power supply of 1.8V, and their dimensions are listed in Table I.

The simulationally measured quantities are presented in Table II. In both filter types – the ladder-based and signal-flow-based – the core filtering parameters remain similar across all three variants. The 3-decibel, -270° , 1dB difference and 1° difference frequencies were maintained, while THD

and noise parameters either stayed the same or improved. Most importantly, however, both transistor area and power consumption saw a meaningful decrease with each optimization step, with only a slight difference between results for both filter types.

For the ladder-based filter, the shared CMFB approach resulted in a 32.4% reduction in power consumption and an 18.9% decrease in transistor area compared to the base variant. Furthermore, incorporating the shared output stage structure reduced power consumption by 42.2% and transistor area by 37.1% compared to the base variant, or by 14.5% and 22.4%, respectively, compared to the shared CMFB variant.

For the signal-flow graph-based filter, a 32.5% reduction in power consumption and a 19.2% decrease in transistor area were achieved by employing the shared CMFB structure. Moreover, introducing the shared output stage approach reduced power consumption by 37.6% and transistor area by 42.4% compared to the base variant. Relative to the shared CMFB variant, these reductions were 14.6% and 22.7%, respectively.

TABLE I.
TRANSISTOR DIMENSIONS IN OTA

Dimensions	Transistors						
	M1-2	M3	M4-5	M6-7	M8-9	M10.0 M10.1	M11.0 M11.1
W [μm]	1.8	2	2	2	0.9	0.9	0.9
L [μm]	0.9	0.6	0.6	0.6	0.6	0.6	0.6
m^a	1	2	1	1	1	1	1

^a. transistor multiplier

V. CONCLUSIONS

This paper proposed an optimized design methodology for OTA-C filters, focusing on reducing power consumption and transistor area while maintaining high performance. Two filter types were analyzed: ladder-based and signal-flow-based designs. Both were implemented in three configurations: the base variant with full OTAs, a variant with shared CMFB circuits, and a fully optimized variant with shared output stages and CMFB circuits.

Simulation results demonstrated significant improvements in resource efficiency achieved without worsening the performance. For ladder-based filters, the shared CMFB structure reduced power consumption by 32.4% and transistor area by 18.9%, while the shared output stage further improved these metrics to 42.2% and 37.1%, respectively, compared to the base variant. Similarly, in signal-flow-based filters, the shared CMFB approach achieved reductions of 32.5% and 19.2%, and the shared output stage, compared to the base variant, yielded reductions of 37.6% and 42.4%, respectively.

Despite challenges such as change of OTA's transconductance due to shared output stages, after necessary modifications the proposed designs successfully maintained key filter characteristics, including frequency, noise and distortion characteristics. These findings highlight the potential of shared sub-blocks in OTA-C filters for achieving high integration, low costs, scalability, and energy efficiency in modern electronic systems.

TABLE II.
THE DESIGNED VI-TH ORDER FILTERS' SIMULATED PARAMETERS

Measured parameters	Ladder-based OTA-C filter			Signal-flow-based OTA-C filter		
	<i>Full OTA</i>	<i>Shared CMFB</i>	<i>Shared CMFB and output stage</i>	<i>Full OTA</i>	<i>Shared CMFB</i>	<i>Shared CMFB and output stage</i>
3dB frequency [MHz]	4.504	4.504	4.513	4.512	4.51	4.501
-270° frequency [MHz]	4.489	4.49	4.489	4.492	4.499	4.489
Frequency of 1dB difference from ideal filter [MHz]	142.7	150.6	43.46	295.5	327.5	46.99
Frequency of 1° difference from ideal filter	6.856	6.867	5.139	6.19	10.34	9.654
Input voltage amplitude for 1% THD [mV]	163.28	163.28	179.1	168	168	184
Input referred noise [mV RMS]	5.717	5.717	4.922	5.189	5.131	4.443
Dynamic range [dB]	26.105	26.105	28.209	27.194	27.292	29.333
Power consumption [μ W]	1006.74	680.76	582.12	890.5	600.8	513
Transistor area [μm^2]	344.82	279.48	216.84	302.94	244.86	189.18

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